GGG-W-643b
April 3, 1963
SUPERSEDING
Fed. Spec. GGG-W-643a
December 9, 1953

FEDERAL SPECIFICATION

# WRENCH, PIPE, INTERNAL-GRIPPING

*This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.*

## 1. SCOPE AND CLASSIFICATION

**1.1 Scope.** This specification covers internal-gripping pipe wrenches for use in installing and removing short and broken nipples or pipe from fittings without marring the exterior finished surface of the pipe and fittings.

1.1.1 *Federal specification coverage.* Federal specifications do not cover all types and classes of the commodities indicated by the titles of the specificatons, but include only those generally used by the Federal Government.

**1.2 Classification.**

1.2.1 *Types, classes and styles.* Internal-gripping pipe wrenches shall be of the following types, classes and styles as specified (see 6.1):

Type I.—Serrated cam.
  Class 1.—For standard pipe only.
  Class 2.—For extra-heavy pipe only.
Type II.—Expansion blade.
  Class 1.—For standard pipe only.
    Style A.—Short body.
    Style B.—Long body.
  Class 2.—For standard and extra-heavy pipe.

## 2. APPLICABLE SPECIFICATIONS AND STANDARDS

**2.1 Specifications and standards.** The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

*Federal Standards*:

Fed. Std. No. 102—Preservation, Packaging, and Packing Levels.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

*Military Specification*:

MIL-H-15424—Hand Tools, Packaging of.

FSC 5120



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*Military Standards:*

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

MIL-STD-130—Identification Marking of U. S. Military Property.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

## 3. REQUIREMENTS

**3.1 Illustrations.** The illustrations shown herein are descriptive and not restrictive and are not intended to preclude the purchase of wrenches otherwise conforming to this specification.

**3.2 Material.** Internal-gripping pipe wrenches shall be of a good quality steel, heat treated to provide wrenches which conform to this specification. Material shall be free of all defects and imperfections which may adversely affect the finished product and shall be such as to produce wrenches capable of long life under hard service conditions.

**3.3 Finish.** Unless otherwise specified (see 6.1), wrenches shall have one of the following finishes:

(*a*) Natural, free from scale, oil treated.
(*b*) Chemically produced oxide coating, oil treated.
(*c*) Chemically produced phosphate coating, oil treated.
(*d*) Zinc or other equally suitable plating.

**3.4** Wrenches shall show no damage or undue distortion when tested as specified in 4.4.2.2.

**3.5 Marking.** Wrenches shall be marked in a plain and permanent manner with the manufacturer's name or with a trademark of such known character that the source of manufacture may be readily determined. They shall also be permanently marked with the pipe size or sizes for which they are adaptable. For military use, marking shall be in accordance with MIL-STD-130, as applicable.

**3.6 Type I, serrated cam.** Type I wrenches shall be similar to figure 1 and shall consist of a body, one end of which shall be designed to accommodate a standard size open-end wrench. The opposite end shall form a circular or semicircular serrated section, concentric with the body. A floating serrated segment or wheel shall be attached to the serrated end of the body so as to create a cam action when force is applied to the wrench end of the body. Wrenches shall be designed to prevent passage of the entire body into the size pipe for which the wrench is intended. The 1-1/2-inch and larger size wrenches may be of welded fabrication.

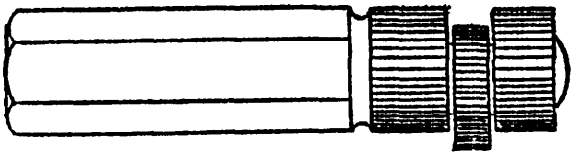


FIGURE 1.—Type I, serrated cam internal-gripping pipe wrench.

**3.6.1** *Hardness.* The shank shall have a Rockwell C hardness of not less than 30 nor more than 45. The serrated sections shall have a Rockwell C hardness of not less than 40 nor more than 52 in sizes larger than 1 inch and not less than 45 nor more than 52 in sizes 1 inch and under.

**3.6.2** *Class I, for standard pipe only.* Class 1 wrenches shall be designed for use with standard pipe sizes as shown in table I.



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Table I - Type I, class 1, serrated cam wrenches for use in standard pipe sizes.

| Nominal pipe size | Diameter serrated section1/ | Length serrated section | Length overall |
|---|---|---|---|
| | | Minimum | Minimum |
| Inches | Inches | Inches | Inches |
| 1/8 | 0.265 | 3/4 | 2-5/8 |
| 1/4 | .350 | 7/8 | 2-5/8 |
| 3/8 | .458 | 1 | 3 |
| 1/2 | .580 | 1 | 3 |
| 3/4 | .770 | 1 | 3 |
| 1 | .987 | 1 | 3 |
| 1-1/4 | 1.275 | 1-3/8 | 3-1/2 |
| 1-1/2 | 1.525 | 1-3/8 | 4 |
| 2 | 1.968 | 1-3/8 | 5 |
| 2-1/2 | 2.375 | 2 | 5 |
| 3 | 2.968 | 2-1/2 | 6 |
| 4 | 3.875 | 3-1/4 | 7-1/2 |

1/ Tolerances.- Diameter of serrated section, 3/8 inch nominal pipe size and below, minus 0.010 inch, plus 0; 1/2 inch to 1-1/4 inches, minus 1/16 inch, plus 0; 1-1/2 inches and over, minus 3/32 inch, plus 0.

3.6.3 *Class 2, for extra-heavy pipe only.* Class 2 wrenches shall be designed for use with extra-heavy pipe sizes as shown in table II.

Table II - Type I, class 2, serrated cam wrenches for use on extra-heavy pipe.

| Nominal pipe size | Diameter, serrated section1/ | Length serrated section | Length overall |
|---|---|---|---|
| | | Minimum | Minimum |
| Inches | Inches | Inches | Inches |
| 1/4 | 0.290 | 7/8 | 2-5/8 |
| 3/8 | .395 | 1 | 3 |
| 1/2 | .505 | 1 | 3 |
| 3/4 | .700 | 1 | 3 |
| 1 | .895 | 1 | 3 |
| 1-1/4 | 1.200 | 1-3/8 | 3-1/2 |
| 1-1/2 | 1.415 | 1-3/8 | 4 |
| 2 | 1.850 | 1-3/8 | 5 |
| 2-1/2 | 2.230 | 2 | 5 |
| 3 | 2.800 | 2-1/2 | 6 |
| 4 | 3.675 | 3-1/4 | 7-1/2 |

1/ Tolerances.- Diameter serrated section, 3/8 inch nominal pipe size and below, minus 0.010 inch, plus 0; 1/2 inch to 1-1/2 inches, minus 1/16 inch, plus 0; 2 inches and over, minus 3/32 inch, plus 0.



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**3.7 Type II, expansion blade.** Type II wrenches shall be similar to figure 2 and shall consist of a body, three or more blades, and an adjusting screw. The body shall be hexagonal on the shank end, of suitable length and cross-sectional dimensions to accomodate a standard size open-end wrench. The remainder of the body shall be cylindrical in shape and slotted to receive the blades. The adjusting screw shall actuate the blade in such manner that the blades shall be concentric with the axis of the body in all positions throughout the adjustment range.

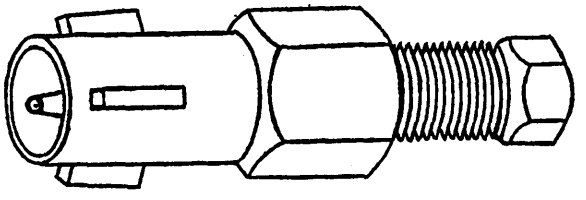


FIGURE 2.—Type II, expansion blade internal-gripping pipe wrench.

**3.7.1** *Hardness.* The body shall have a Rockwell C hardness of not less than 40 nor more than 45, or be casehardened to a minimum depth of 0.020 inch and shall have a hardness of 86 to 90 on the Rockwell 15-N scale. The blades shall have a Rockwell C hardness of not less than 51 nor more than 55. The screw shall have a Rockwell C hardness of 51 to 60 on the C scale or be casehardened to a minimum depth of 0.015 inch and shall have a hardness of 86 to 90 on the Rockwell 15-N scale.

**3.7.2** *Class* 1, *for standard pipe only.* Class 1 wrenches shall be designed for use with standard pipe sizes as shown in tables III and IV, as applicable.

**3.7.2.1** *Style A, short body wrench.* Style A wrenches shall have dimensions as shown in table III.

**Table III - Type II, class 1, style A short body expansion blade wrenches for use with standard pipe only.**

| For nominal pipe size | Length of blades (±1/16 inch) | Overall length of body (±1/4 inch) | Hexagon shank size (across flat) |
|---|---|---|---|
| Inches | Inches | Inches | Inches |
| 1/2 | 39/64 | 2-5/8 | 3/4 |
| 3/4 | 43/64 | 2-5/8 | 1 |
| 1 | 11/16 | 2-5/8 | 1 |
| 1-1/4 to 1-1/2 | 31/32 | 4 | 1-3/8 |
| 2 to 2-1/2 | 31/32 | 4-1/2 | 1-3/4 |



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Table IV - Type II, class 1, style B long body expansion blade wrenches for use with standard pipe only.

| For nominal pipe size | Length of blades. (±1/16 inch) | Overall length of body (±1/4 inch) | Hexagon shank size (across flat) |
|---|---|---|---|
| Inches | Inches | Inches | Inches |
| 1/2 | 39/64 | 5-3/4 | 7/8 |
| 3/4 | 43/64 | 5-3/4 | 1 |
| 1 | 11/16 | 6 | 1 |
| 1-1/4 to 1-1/2 | 31/32 | 9 | 1-3/8 |

Table V - Type II, class 2 expansion blade wrenches for use with both standard and extra-heavy pipe.

| For nominal pipe size | Length of blades (±1/16 inch) | Overall length of body (±1/4 inch) | Hexagon shank size (across flat) |
|---|---|---|---|
| Inches | Inches | Inches | Inches |
| 3 to 3-1/2 | 1-15/32 | 7-5/16 | 2-1/16 |
| 4 to 4-1/2 | 1-15/32 | 7-5/16 | 2-1/2 |
| 5 to 5-1/2 | 1-15/32 | 9-3/4 | 2-1/2 |
| 6 | 1-15/32 | 12-1/2 | 2-1/2 |

3.7.2.2 *Style B, long body wrench.* Style B wrenches shall have dimensions as shown in table IV.

3.7.3 *Class 2, for use with both standard and extra-heavy pipe.* Class 2 wrenches shall be designed for use with both standard and extra-heavy pipe sizes shown in table V.

3.8 **Workmanship.** The workmanship shall be first class throughout. The wrenches shall be free from fins, burrs, rough edges, or any other defect which may impair the serviceability or durability of the tool.

## 4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 **Sampling for quality conformance inspection.**



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Table VI - Sampling for examination
AQL (approx.) = 2.5 percent defective.

| Number of wrenches in inspection lot | Number of wrenches in sample | Acceptance number (defectives) | Rejection number (defectives) |
|---|---|---|---|
| 15 and under | 7 | 0 | 1 |
| 16 to 40 | 10 | 0 | 1 |
| 41 to 110 | 15 | 1 | 2 |
| 111 to 300 | 25 | 1 | 2 |
| 301 to 500 | 35 | 2 | 3 |
| 501 to 800 | 50 | 3 | 4 |
| 801 to 1,300 | 75 | 4 | 5 |
| 1,301 and over | 110 | 6 | 7 |

4.2.1 *Lot.* All wrenches of the same type, class, style, and size shall be considered a lot for the purpose of quality conformance inspection.

4.2.2 *Sampling for examination.*

4.2.2.1 *Sampling for intermittent procurement in isolated lots.* A random sample of wrenches from each inspection lot of material shall be examined for conformance with this specification, with lot acceptance based on the single sampling inspection requirements in accordance with table VI and the procedures of MIL-STD-105.

4.2.2.2 *Sampling for continuous production.* Where there is continuous production and inspection lots include at least 200 items, sampling for examination shall be conducted in accordance with MIL-STD-105 at inspection level I and AQL = 2.5 percent defective.

4.2.3 *Sampling for tests.* A random sample of wrenches shall be selected from each lot in accordance with inspection level L-7 of MIL-STD-105, and shall be subjected to the tests specified in 4.4. If any wrench fails in any test, the lot represented shall be rejected.

4.3 **Examination.** Each of the sample wrenches selected in accordance with 4.2.2 shall be examined to verify compliance with this specification. Defects shall be classified in accordance with table VII. Any wrench in the sample containing one or more defects shall be rejected, and if the number of defective wrenches in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

Table VII - Classification of defects in accordance with MIL-STD-105.

| Categories | Defects |
|---|---|
| Critical: | None defined |
| Major: | |
| 101 | Type, class, and size (for nominal pipe size) not as specified. |
| 102 | Material nonconforming; not free from cracks, seams, splits, laps, scale, or corrosion. |
| 103 | Finish not smooth, not free from fins, burrs, rough, or sharp exposed edges. |
| 104 | Wrench not coated or treated as specified; coating when applicable not smooth, continuous, uniform, or adherent, not free from uncoated areas, pits, blisters, or nodules. |
| 105 | Incomplete, parts missing, damaged, inoperable, or not as designated. |
| 106 | Working sections not serrated or floating section not securely attached for proper cam action (type I only). |
| 107 | Diameter of serrated section not within allowable limits, serrated section and overall lengths less than the minimum limits (type I only). |
| 108 | Cross section (shape) of body not as required (type I only). |
| 109 | Number of blades less than required; blades not concentric with axis of body for all positions in the adjustment range (type II only). |
| 110 | Length of blades, overall length and shank size across flats not within allowable limits (type II only). |
| 111 | Adjusting screw sticks, binds or threads crossed, stripped, cracked, or torn (type II only). |
| 112 | Shank end cross section (shape) nonconforming, not of proper length (type II only). |
| Minor: | |
| 201 | Marking, manufacturer's name or trademark and pipe size(s) missing, incorrect, illegible, or not permanent. |



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4.4 **Test procedures.**

4.4.1 *Hardness.* Each of the sample wrenches selected in accordance with 4.2.3 shall be tested to determine conformance with 3.6.1 and 3.7.1. Two or more determinations of the hardness shall be made on the specified portions of the tool. All values shall fall within the hardness range specified.

4.4.2 *Performance tests.*

4.4.2.1 Each sample wrench selected in accordance with 4.2.3 shall be tested with the maximum-size pipe for which it is adaptable.

4.4.2.2 Using standard-practice tightening methods, a standard pipe nipple shall be set up in a standard pipe fitting with copalite, shellac, or litharge and glycerin and shall be allowed to dry until the pipe dope is firmly set. The wrench shall then be used to remove the nipple from the fitting and shall show no damage to the wrench and no undue distortion of the fitting (see 3.4).

4.5 **Inspection of preparation for delivery.** Sample wrenches and packages shall be selected and inspected to verify conformance to the requirements of section 5 herein.

## 5. PREPARATION FOR DELIVERY

(Civil agencies should refer to Fed. Std. No. 102 for definitions and applications of the various levels of packaging protection for supplies and equipment.)

5.1 **Cleaning, preservation, and packaging.**

5.1.1 *Level A.* Wrenches shall be preserved and packaged in accordance with level A of MIL-H-15424.

5.1.2 *Level C.* Cleaning, preservation, and packaging shall be in accordance with the contractor's commercial practice.

5.2 **Packing.**

5.2.1 *Level A.* Wrenches packaged as specified (see 6.1), shall be packed in accordance with level A requirements of MIL-H-15424.

5.2.2 *Level B.* Wrenches, packaged as specified (see 6.1), shall be packed in accordance with level B requirements of MIL-H-15424.

5.2.3 *Level C.* Wrenches, packaged as specified (see 6.1), shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with the rules or regulations as applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Civil agencies.* In addition to any special marking required by the contract or order, interior and exterior shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.3.2 *Military agencies.* In addition to any special marking required by the contract or order, interior and exterior shipping containers shall be marked in accordance with MIL-STD-129.

## 6. NOTES

6.1 **Ordering data.** Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, number, and date of this specification.

(*b*) Type, class, style, and size required (see 1.2 and tables I to V, inclusive).

(*c*) Finish, if different (see 3.3).

(*d*) Applicable levels of packaging and packing required (see section 5).

(*e*) Marking, if different (see 5.3).



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6.2 **Transportation description.** Transportation description applicable to these items is:

Wrenches, not otherwise indexed by name, without attachments.
Carload weight 30,000 pounds.
Truckload weight 30,000 pounds.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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